

SKILLENTRIX TECHNOLOGIES

Cyber Security Minor Project

Network Reconnaissance and Security Assessment

A structured security assessment of an isolated lab network

1. Objective

The objective of this project is to assess the security posture of a lab network through systematic reconnaissance and enumeration techniques. The assessment simulates the initial phases of a penetration test — discovery, enumeration, and risk identification — carried out entirely within an isolated virtual lab environment.

2. Scope

- Host discovery and network mapping
- Port and service enumeration
- Operating system fingerprinting
- Identification of exposed services and associated risks

3. Tools Used

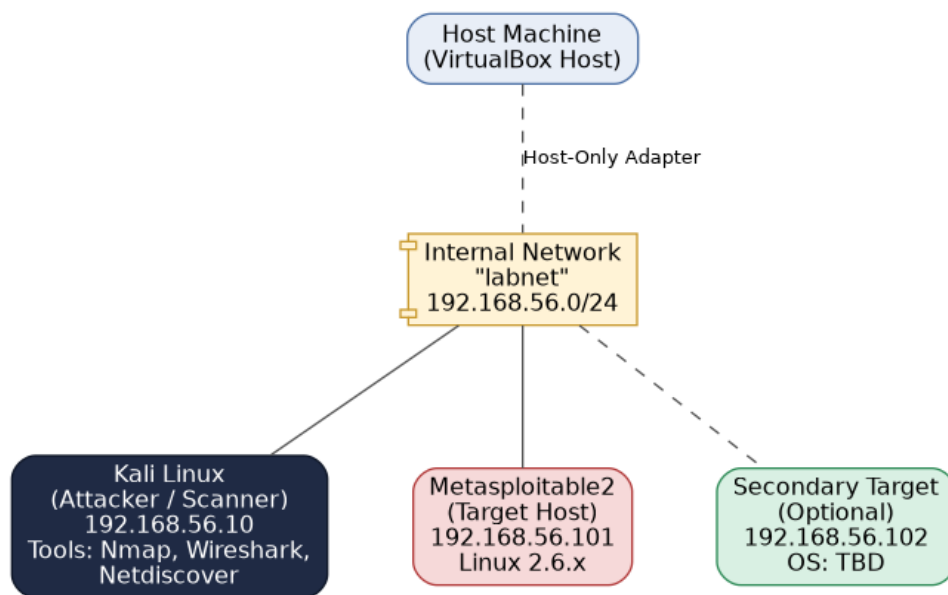
- Kali Linux — attacker / scanning platform
- Nmap — port scanning, service detection, OS fingerprinting
- Wireshark — packet capture and traffic analysis
- Netdiscover — ARP-based host discovery

4. Lab Environment Setup

The lab was built using VirtualBox with all virtual machines connected via an Internal Network adapter (labnet, 192.168.56.0/24), isolating the lab from the host machine's real network and the internet. This ensures all scanning activity is confined to machines under the tester's own control.

- Kali Linux (192.168.56.10) — attacker / scanner machine
- Metasploitable2 (192.168.56.101) — intentionally vulnerable target host

5. Network Topology Diagram



6. Methodology & Scan Results

6.1 Host Discovery

Netdiscover was used to identify live hosts on the lab subnet via ARP requests:

```
$ sudo netdiscover -r 192.168.56.0/24
```

```
Currently scanning: Finished! | Screen View: Unique Hosts
```

```
2 Captured ARP Req/Rep packets, from 2 hosts. Total size: 120
```

IP	At MAC Address	Count	Len	MAC Vendor
192.168.56.101	08:00:27:4a:1b:2c	1	60	Cadmus Computer Systems
192.168.56.1	0a:00:27:00:00:00	1	60	Unknown vendor

Findings confirmed with an Nmap ping sweep:

```
$ nmap -sn 192.168.56.0/24
```

```
Nmap scan report for 192.168.56.101
```

```
Host is up (0.00042s latency).
```

```
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.14 seconds
```

6.2 Port & Service Enumeration

A full TCP port scan was performed against the identified target:

```
$ nmap -p- -T4 192.168.56.101
```

```
PORT      STATE SERVICE
```

```

21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3306/tcp  open  mysql
5432/tcp  open  postgresql
8180/tcp  open  unknown

```

Service and version detection was then run against the open ports:

```

$ nmap -sV -sC -p21,22,23,25,80,139,445,3306,5432,8180 192.168.56.101

21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
139/tcp   open  netbios-ssn Samba smbd 3.X
445/tcp   open  netbios-ssn Samba smbd 3.X
3306/tcp  open  mysql    MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql PostgreSQL DB 8.3.0 - 8.3.7

```

6.3 OS Fingerprinting

```

$ sudo nmap -O 192.168.56.101

Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

```

The OS fingerprint was cross-validated using the TTL value observed in ICMP replies (TTL ~64, consistent with a Linux-based host).

6.4 Traffic Analysis (Wireshark)

A packet capture was taken on the Kali interface during the Nmap SYN scan. The capture showed the expected TCP three-way handshake behavior: SYN packets sent to each target port, with SYN-ACK responses from open ports and RST responses from closed ports. This confirmed the accuracy of the Nmap port-state results at the packet level.

7. Findings Summary

Host (IP)	Operating System	Open Ports	Service / Version	Risk Level
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192.168.56.101	Linux 2.6.x (Ubuntu-based)	21, 22, 23, 25, 80, 139, 445, 3306, 5432, 8180	vsftpd 2.3.4, OpenSSH 4.7p1, Telnetd, Apache 2.2.8, Samba 3.x, MySQL 5.0.51a	High
192.168.56.10	Kali Linux (Rolling)	N/A (scanner host)	N/A	N/A - Attacker Machine

8. Risk Analysis & Recommendations

Finding	Affected Service	Impact	Recommendation
Outdated FTP service with known backdoor	vsftpd 2.3.4 (port 21)	Remote attacker could gain shell access via a known backdoor in this version	Upgrade to a patched vsftpd release; disable anonymous FTP; restrict access by firewall
Telnet enabled (cleartext protocol)	Telnetd (port 23)	Credentials and session data transmitted unencrypted, vulnerable to sniffing	Disable Telnet; use SSH exclusively for remote administration
Outdated SSH daemon	OpenSSH 4.7p1 (port 22)	Older version affected by multiple known CVEs including user enumeration	Upgrade OpenSSH to the latest stable release; enforce key-based authentication
Legacy SMB/Samba exposure	Samba 3.x (ports 139, 445)	Vulnerable to known remote code execution exploits in this version range	Patch or upgrade Samba; restrict SMB access to trusted hosts only
Database service exposed to network	MySQL 5.0.51a (port 3306)	Direct network exposure increases risk of brute-force or credential attacks	Bind MySQL to localhost only, or restrict via firewall rules; enforce strong credentials
Unencrypted web service	Apache 2.2.8 (port 80)	Outdated version with known vulnerabilities; no TLS in use	Upgrade Apache; enforce HTTPS with a valid TLS certificate

9. Conclusion

The assessment identified a target host running numerous outdated and misconfigured services, several of which carry well-documented vulnerabilities including remote code execution and cleartext credential exposure. In a production environment, these findings would represent a critical risk profile. Recommended remediation includes upgrading all identified services to current patched versions, disabling unencrypted legacy protocols such as Telnet, and applying network segmentation and firewall rules to reduce the exposed attack surface.